

Designing Interactive Chart Learning Tools with Generative AI for Financial Education in Real-Time Investment

Meiying Li
College of Design and Innovation
Tongji University
Shanghai, China
mzlzyca@tongji.edu.cn

Yuting Xie
College of Design and Innovation
Tongji University
Shanghai, China
yukting@tongji.edu.cn

Jiayu Pan
College of Design and Innovation
Tongji University
Shanghai, China
pracless@tongji.edu.cn

Qi Wang
College of Design and Innovation
Tongji University
Shanghai, China
qiwangdesign@tongji.edu.cn

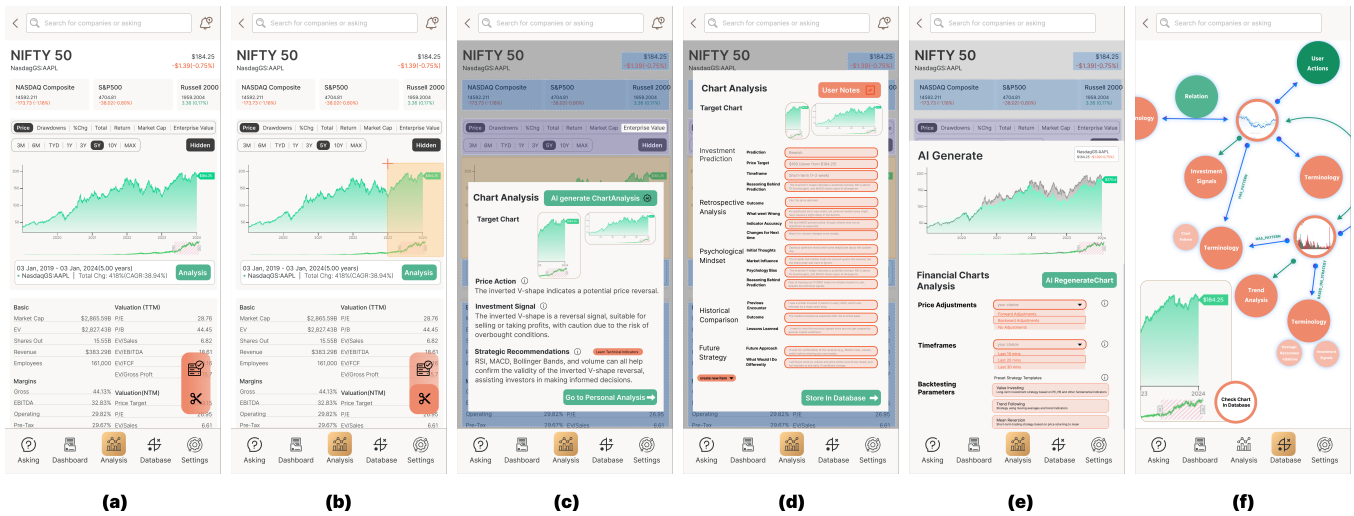


Figure 1: (a) shows a typical mobile investment interface. (b) Our tool enables users to capture charts directly from the trading page. (c), (d), and (f) illustrate how generative AI analyzes captured charts to extract technical terminology (e.g., chart patterns) and generate investment strategy suggestions. Users can annotate charts and reflect on their decisions, with all information stored in a database for future review. (c) and (e) demonstrate how the system converts chart data into Vega-Lite specifications, allowing users to modify inputs and regenerate charts for interactive learning. (e) highlights the database storage of all chart-related information.

Abstract

Digital investment platforms now provide widespread access to financial services, but a significant gap remains between platform usability and user comprehension. Our questionnaire survey reveals that many novice investors struggle to grasp the complex financial knowledge embedded in investment tools, as charts often serve as educational resources but remain difficult to understand. To address

this challenge, we conducted a survey examining how investors utilize and learn from charts during investment and developed an interactive financial education tool that employs generative AI to interpret financial charts, enabling users to acquire investment knowledge within their actual decision-making contexts.

CCS Concepts

• **Human-centered computing** → *User interface toolkits.*

Keywords

Financial Education, Chart Interpretation, Generative AI, Interactive Learning, Financial Knowledge Graphs

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UIST Adjunct '25, Busan, Republic of Korea

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ACM ISBN 979-8-4007-2036-9/25/09

<https://doi.org/10.1145/3746058.3758384>

ACM Reference Format:

Meiying Li, Yuting Xie, Jiayu Pan, and Qi Wang. 2025. Designing Interactive Chart Learning Tools with Generative AI for Financial Education in Real-Time Investment. In *The 38th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '25), September 28–October 01, 2025, Busan, Republic of Korea*. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3746058.3758384>

1 Introduction

With the rapid development of digital finance, ubiquitous digital devices have made accessing online financial services incredibly convenient, reducing barriers to individual fund management [6]. However, many potential investors lack necessary financial literacy to effectively navigate modern financial systems, indicating a strong need for investment education [3, 13]. Recent studies have highlighted the importance of financial knowledge learning in investment education [1, 4, 5, 10].

While financial charts serve as both investment tools and effective mediums for conveying financial concepts [7, 14], users face challenges understanding these charts due to their complex graphical and textual information. A user study with 12 novice investors showed that participants struggled to interpret financial charts and terminology, revealing a gap between interface design and user learning needs.

With generative AI advancement, studies have highlighted its potential in financial investment education [2, 12]. Recent research shows AI applications can extract complex financial data from charts [8, 11] and enhance understanding [9]. However, these studies emphasize investment outcomes while overlooking the knowledge acquisition process.

To address this gap, we propose an interactive prototype that integrates real-time chart capture, AI-assisted analysis, and personalized knowledge construction. By leveraging generative AI, the system extracts financial knowledge from charts and presents it through modifiable visualizations, supporting knowledge internalization and strategy development in real investment contexts.

2 Design Requirements from User Survey

We conducted a survey with 6 intermediate investors (2-5 years of experience) about how they utilize and learn from charts during investment, and summarized the following design requirements.

Chart Capture and Management The system should capture market charts in real-time and build a personal repository for analysis and reflection, as users frequently screenshot charts and maintain investment notes during trading. It should store relationships among chart patterns, technical terms, strategies, and user behavior in a graph database, enabling users to recognize familiar patterns and leverage insights across different market scenarios.

Chart Annotation The system should employ generative AI to provide initial annotations across pattern recognition, technical signal interpretation, and strategic recommendations. Users can edit and extend these annotations to create personalized knowledge entries while recording investment decisions and thought processes identified in our survey.

Interactive Learning The system should leverage interactive learning to improve chart comprehension by converting complex data into structured information through chart-text integration.

The design should integrate these methods into actual investment decision-making processes.

3 User Workflow

In a typical use case, an investor interacts with a mobile trading interface (Figure 1a) and captures a chart directly from the trading page using our tool (Figure 1b). The system applies generative AI to analyze the captured chart, extract technical terminology (e.g., chart patterns), and generate corresponding investment strategy suggestions (Figure 1c, d, f). Users can annotate the chart and record reflections on their decision-making process, with all information stored in a database for future review (Figure 1f). Additionally, the system converts chart data into Vega-Lite specifications (Figure 1c, e), enabling users to adjust input parameters and regenerate charts to support interactive learning. The database storage of chart-related information is further illustrated in Figure 1e.

4 System Implementation

Our system comprises three core components: an encoding module responsible for parsing and interpreting financial charts using generative AI, a decoding module that presents the processed knowledge through an interactive interface, and a financial knowledge graph that tracks and contextualizes user learning. The frontend uses React¹ for responsive design, integrating html2canvas² for chart capturing. The backend employs Node.js for RESTful services and OpenAI GPT-4o API³ for chart pattern analysis and strategy generation. Data storage combines PostgreSQL for chart structure and Neo4j⁴ graph database for modeling relationships between charts, terms, strategies, and user decision paths.

The encoding module leverages GPT-4o with scaffolded prompting to extract key features including price actions, technical indicators, and strategy signals with accompanying explanatory text. The output is structured in both JSON format for visualization and interaction, and a graph-based schema for long-term knowledge representation. This dual-format structure captures entities (e.g., candlestick patterns, indicators), relationships (e.g., strategy connections), and events (e.g., trade execution) to support multi-level financial understanding.

The decoding module renders JSON-encoded information into interactive charts using Vega-Lite specification⁵. Users can modify chart configurations (price adjustments, timeframes, backtesting parameters) through interface interactions, triggering the system to dynamically update the JSON schema and generate new Vega-Lite specifications via generative AI. These interactions support incremental learning while user actions are recorded for interface adaptation and knowledge tracking.

The knowledge graph built in Neo4j stores data collected from AI chart analysis and organizes user-acquired investment knowledge. Key node types include Chart (capturing chart images and timestamps), Pattern (e.g., flags, wedges), Term (e.g., Bollinger Bands, golden cross), Strategy (LLM or user-generated tactics), and

¹<https://react.dev>

²<https://github.com/vre2h/use-react-screenshot/blob/master/README.md>

³<https://openai.com/api/>

⁴<https://neo4j.com/>

⁵<https://vega.github.io/vega-lite/>

User Action (user decisions and outcomes). These entities are connected through domain-specific relations such as *HAS_PATTERN*, *USES_TERM*, *BASED_ON_STRATEGY*, and *DECIDED_BY_USER*. The system supports AI-assisted querying: when a new chart is captured, it retrieves similar historical patterns based on image hashes or structural features, and returns related strategies and decision outcomes to support user reflection and decision-making.

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